

To show that the series $\sum_{k=1}^{\infty} \frac{1}{k}$ is divergent, we will prove that

$$\sum_{k=1}^n \frac{1}{k} > \ln(n) + \frac{1}{n}, \quad \forall n \geq 1. \quad (3.11)$$

Since $\frac{1}{x}$ is a decreasing function, it follows that

$$\frac{1}{x} < \frac{1}{k}, \quad \forall k < x < k+1.$$

Then,

$$\begin{aligned} \int_1^n \frac{1}{x} dx &= \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{x} dx \\ &< \sum_{k=1}^{n-1} \int_k^{k+1} \frac{1}{k} dx \\ &= \sum_{k=1}^{n-1} \frac{1}{k} \\ &= -\frac{1}{n} + \sum_{k=1}^n \frac{1}{k}. \end{aligned} \quad (3.12)$$

From (3.12), we find that

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k} &> \int_1^n \frac{1}{x} dx + \frac{1}{n} \\ &= \ln(n) + \frac{1}{n}, \quad \forall n \geq 1, \end{aligned}$$

which is what we wanted to prove; see (3.11).

Similarly, note that

$$\frac{1}{x \ln(x)} < \frac{1}{k \ln(k)}, \quad \forall k < x < k+1,$$

and therefore

$$\begin{aligned}
 \int_2^{n+1} \frac{1}{x \ln(x)} dx &= \sum_{k=2}^n \int_k^{k+1} \frac{1}{x \ln(x)} dx \\
 &< \sum_{k=2}^n \int_k^{k+1} \frac{1}{k \ln(k)} dx \\
 &= \sum_{k=2}^n \frac{1}{k \ln(k)}. \quad (3.13)
 \end{aligned}$$

Since

$$\int_2^{n+1} \frac{1}{x \ln(x)} dx = \ln(\ln(n+1)) - \ln(\ln(2)),$$

we obtain from (3.13) that

$$\sum_{k=2}^n \frac{1}{k \ln(k)} > \ln(\ln(n+1)) - \ln(\ln(2)),$$

and we conclude that the series $\sum_{k=2}^{\infty} \frac{1}{k \ln(k)}$ is divergent.

Note: Although not needed to answer this question, it can be shown that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6},$$

and

$$\lim_{n \rightarrow \infty} \left(\sum_{k=1}^{\infty} \frac{1}{k} - \ln(n) \right) = \gamma,$$

where $\gamma \approx 0.57721$ is Euler's constant. $\square$

Question 9. Compute

$$\int \frac{1}{1+x^2} dx.$$

Answer: Use the substitution $x = \tan(z)$. Then, $dx = \frac{1}{\cos^2(z)} dz$ and

$$\begin{aligned}\int \frac{1}{1+x^2} dx &= \int \frac{1}{(1+\tan^2(z)) \cos^2(z)} dz \\ &= \int \frac{1}{\cos^2(z) + \sin^2(z)} dz \\ &= \int 1 dz \\ &= z + C,\end{aligned}$$

where C is a real constant, since $\cos^2(z) + \sin^2(z) = 1$ for any z . Solving $x = \tan(z)$ for z , we obtain that $z = \arctan(x)$, and therefore

$$\int \frac{1}{1+x^2} dx = \arctan(x) + C. \quad \square$$

Question 10. Compute

$$\int x \ln(x) dx \quad \text{and} \quad \int x e^x dx.$$

Answer: By integration by parts,

$$\begin{aligned}\int x \ln(x) dx &= \frac{x^2}{2} \ln(x) - \int \frac{x^2}{2} \cdot \frac{1}{x} dx \\ &= \frac{x^2}{2} \ln(x) - \frac{1}{2} \int x dx \\ &= \frac{x^2 \ln(x)}{2} - \frac{x^2}{4} + C; \\ \int x e^x dx &= x e^x - \int 1 \cdot e^x dx \\ &= x e^x - e^x + C. \quad \square\end{aligned}$$

Question 11. Compute

$$\int x^n \ln(x) \, dx.$$

Answer: If $n \neq -1$, we use integration by parts and find that

$$\begin{aligned} \int x^n \ln(x) \, dx &= \frac{x^{n+1}}{n+1} \ln(x) - \int \frac{x^{n+1}}{n+1} \cdot \frac{1}{x} \, dx \\ &= \frac{x^{n+1} \ln(x)}{n+1} - \frac{1}{n+1} \int x^n \, dx \\ &= \frac{x^{n+1} \ln(x)}{n+1} - \frac{x^{n+1}}{(n+1)^2} + C. \end{aligned}$$

For $n = -1$, we obtain that

$$\int \frac{\ln(x)}{x} \, dx = \frac{(\ln(x))^2}{2} + C,$$

where C is a real constant, since

$$\left(\frac{(\ln(x))^2}{2} \right)' = \frac{1}{2} \cdot 2 \ln(x) \cdot (\ln(x))' = \frac{\ln(x)}{x}. \quad \square$$

Question 12. Compute

$$\int (\ln(x))^n \, dx.$$

Answer: For every integer $n \geq 0$, let

$$f_n(x) = \int (\ln(x))^n \, dx.$$

By using integration by parts, we find that, for any $n \geq 1$,

$$\begin{aligned}
 & \int (\ln(x))^n dx \\
 &= x(\ln(x))^n - \int x ((\ln(x))^n)' dx \\
 &= x(\ln(x))^n - \int x \cdot n(\ln(x))^{n-1} \cdot (\ln(x))' dx \\
 &= x(\ln(x))^n - \int x \cdot n(\ln(x))^{n-1} \cdot \frac{1}{x} dx \\
 &= x(\ln(x))^n - n \int (\ln(x))^{n-1} dx,
 \end{aligned}$$

and therefore

$$f_n(x) = x(\ln(x))^n - n f_{n-1}(x), \quad \forall n \geq 1. \quad (3.14)$$

Note that

$$f_0(x) = \int 1 dx = x + C.$$

Thus, the recursion (3.14) can be used to find the values of $f_n(x)$ for all n . For example:

$$\begin{aligned}
 f_1(x) &= x \ln(x) - f_0(x) = x(\ln(x) - 1) + C; \\
 f_2(x) &= x(\ln(x))^2 - 2f_1(x) \\
 &= x((\ln(x))^2 - 2\ln(x) + 2) + C.
 \end{aligned}$$

The following general formula can be obtained by induction:

$$\int (\ln(x))^n dx = x \sum_{k=0}^n \frac{(-1)^{n-k} n!}{k!} (\ln(x))^k + C,$$

for all $n \geq 0$. $\square$

Question 13. Solve the ODE

$$y'' - 4y' + 4y = 1. \quad (3.15)$$

Answer: Note that (3.15) is a second order non homogeneous linear ODE with constant coefficients. The homogeneous ODE associated to (3.15) is

$$y'' - 4y' + 4y = 0, \quad (3.16)$$

whose characteristic equation, $z^2 - 4z + 4 = 0$, has a double root $z_1 = z_2 = 2$. Thus, the solution to the homogeneous ODE (3.16) is

$$y(x) = c_1 e^{2x} + c_2 x e^{2x}, \quad (3.17)$$

where c_1 and c_2 are constants.

Since the constant function $y_0(x) = \frac{1}{4}$ is a solution to the non homogeneous ODE (3.15), we conclude from (3.17) that the general form of the solution of (3.15) is

$$y(x) = c_1 e^{2x} + c_2 x e^{2x} + \frac{1}{4}. \quad \square$$

Question 14. Find $f(x)$ such that

$$f'(x) = f(x)(1 - f(x)). \quad (3.18)$$

Answer: Note that (3.18) is as an ODE with separable variables and can be written as follows:

$$\frac{y'}{y(1-y)} = 1, \quad (3.19)$$

where $y = f(x)$. By integrating (3.19) with respect to x we obtain that

$$\int \frac{y'}{y(1-y)} dx = \int 1 dx = x + C_1, \quad (3.20)$$

where $C_1 \in \mathbb{R}$ is a real constant.

Note that $dy = y'dx$, and therefore

$$\begin{aligned}
 \int \frac{y'}{y(1-y)} dx &= \int \frac{1}{y(1-y)} dy \\
 &= \int \left(\frac{1}{y} + \frac{1}{1-y} \right) dy \\
 &= \int \frac{1}{y} dy + \int \frac{1}{1-y} dy \\
 &= \ln(|y|) - \ln(|1-y|) \\
 &= \ln \left| \frac{y}{1-y} \right|. \tag{3.21}
 \end{aligned}$$

From (3.20) and (3.21), it follows that

$$\ln \left| \frac{y}{1-y} \right| = x + C_1,$$

and therefore

$$\left| \frac{y}{1-y} \right| = e^{x+C_1} = C_2 e^x,$$

where $C_2 = e^{C_1} > 0$ is a positive real constant.

Thus, either $\frac{y}{1-y} = C_2 e^x$, or $\frac{y}{1-y} = -C_2 e^x$, which can be written as

$$\frac{y}{1-y} = C e^x, \tag{3.22}$$

where C is a real constant.

From (3.22), we obtain that $y = \frac{C e^x}{1 + C e^x}$. We conclude that the ODE (3.18) has the following solution:

$$f(x) = \frac{C e^x}{1 + C e^x},$$

where $C \in \mathbb{R}$ is a fixed constant. $\square$

Question 15. Derive the Black-Scholes PDE.

Answer: Consider an asset with spot price S following a lognormal distribution with drift μ and volatility σ and paying dividends continuously at rate q . Then,

$$dS = (\mu - q)Sdt + \sigma SdW_t,$$

where W_t , $t \geq 0$, is a Wiener process.

Let $V = V(S, t)$ be the value at time t of a replicable non path dependent derivative security on this asset, when the underlying is priced at S . Set up a portfolio Π made of a long position in the derivative security V and a short position in

$$\Delta = \frac{\partial V}{\partial S} \quad (3.23)$$

units of the asset. Then,

$$\Pi = V - \Delta S.$$

Denote by dS , dV , and $d\Pi$ the changes in the values of S , V , and Π , respectively, over an infinitesimally small time period dt . Then,

$$d\Pi = dV - \Delta dS - \Delta qSdt, \quad (3.24)$$

where $\Delta qSdt$ is the dividend payment owed over the time dt on the short Δ units asset position. From (3.23) and (3.24), we find that

$$d\Pi = dV - \frac{\partial V}{\partial S}dS - qS\frac{\partial V}{\partial S}dt. \quad (3.25)$$

From Itô's formula, it follows that

$$\begin{aligned} dV &= \left(\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} \right) dt \\ &\quad + \frac{\partial V}{\partial S}dS, \end{aligned} \quad (3.26)$$

and, from (3.25) and (3.26), we obtain that

$$d\Pi = \left(\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} - qS\frac{\partial V}{\partial S} \right) dt, \quad (3.27)$$

which means that the value of the portfolio Π is deterministic over the small time period dt . For no-arbitrage, the value of the portfolio Π must grow at the risk-free rate over the time period dt , i.e., $d\Pi = r\Pi dt$, where r denotes the risk-free rate. Thus,

$$\begin{aligned} d\Pi &= r\Pi dt = r(V - \Delta S)dt \\ &= \left(rV - rS \frac{\partial V}{\partial S} \right) dt. \end{aligned} \quad (3.28)$$

From (3.27) and (3.28), we find that

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} - qS \frac{\partial V}{\partial S} = rV - rS \frac{\partial V}{\partial S},$$

and therefore

$$\frac{\partial V}{\partial t} + \frac{\sigma^2 S^2}{2} \frac{\partial^2 V}{\partial S^2} + (r - q)S \frac{\partial V}{\partial S} - rV = 0,$$

which is the Black-Scholes PDE for $V(S, t)$. $\square$

3.2 Covariance and correlation matrices. Linear algebra.

Question 1. Show that any covariance matrix is symmetric positive semidefinite. Show that the same is true for correlation matrices.

Answer: Let Σ_X and Ω_X be the covariance matrix and the correlation matrix of n random variables $X_1, X_2, \dots, X_n$. It is easy to see that Σ_X and Ω_X are symmetric matrices:

$$\begin{aligned}\Sigma_X(j, k) &= \text{cov}(X_j, X_k) = \text{cov}(X_k, X_j) \\ &= \Sigma_X(k, j), \quad \forall 1 \leq j, k \leq n; \\ \Omega_X(j, k) &= \text{corr}(X_j, X_k) = \text{corr}(X_k, X_j) \\ &= \Omega_X(k, j), \quad \forall 1 \leq j, k \leq n.\end{aligned}$$

Let $c_1, c_2, \dots, c_n$ be real numbers, and let $C = (c_i)_{i=1:n}$ be a column vector of size n . Recall that

$$\text{var} \left(\sum_{i=1}^n c_i X_i \right) = C^t \Sigma_X C. \quad (3.29)$$

Since the variance of any random variable is nonnegative, it follows that

$$C^t \Sigma_X C \geq 0, \quad \forall C \in \mathbb{R}^n, \quad (3.30)$$

and we conclude that Σ_X is a symmetric positive semidefinite matrix.

For completeness, we include a proof of (3.29) here.

Let $Y = \sum_{i=1}^n c_i X_i$. Then,

$$Y - E[Y] = \sum_{i=1}^n c_i (X_i - \mu_i),$$

where $\mu_i = E[X_i]$, for $i = 1 : n$, and therefore

$$\begin{aligned} & \text{var} \left(\sum_{i=1}^n c_i X_i \right) \\ &= \text{var}(Y) = E[(Y - E[Y])^2] \\ &= E \left[\left(\sum_{i=1}^n c_i (X_i - \mu_i) \right)^2 \right] \end{aligned} \quad (3.31)$$

$$\begin{aligned} &= E \left[\sum_{1 \leq j, k \leq n} c_j c_k (X_j - \mu_j)(X_k - \mu_k) \right] \\ &= \sum_{1 \leq j, k \leq n} c_j c_k E[(X_j - \mu_j)(X_k - \mu_k)] \\ &= \sum_{1 \leq j, k \leq n} c_j c_k \text{cov}(X_j, X_k) \\ &= \sum_{1 \leq j, k \leq n} c_j c_k \Sigma_X(j, k) \\ &= C^t \Sigma_X C, \end{aligned} \quad (3.32)$$

where, for (3.31) and for (3.32), we used the following facts, respectively:

$$\begin{aligned} \left(\sum_{i=1}^n z_i \right)^2 &= \sum_{1 \leq j, k \leq n} z_j z_k, \quad \forall z_i \in \mathbb{R}, \quad i = 1 : n; \\ x^t A x &= \sum_{1 \leq j, k \leq n} x_j x_k A(j, k), \end{aligned}$$

for any $n \times 1$ vector $x = (x_i)_{i=1:n} \in \mathbb{R}^n$, and for any $n \times n$ matrix A .

To show that Ω_X is a symmetric positive semidefinite matrix, recall the following correspondence between covariance matrices and correlation matrices:

$$\Sigma_X = D_{\sigma_X} \Omega_X D_{\sigma_X}, \quad (3.33)$$

where $D_{\sigma_X} = \text{diag}(\sigma_i)_{i=1:n}$ is the diagonal matrix with entries equal to the standard deviations of the n random variables, i.e., $\sigma_i^2 = \text{var}(X_i)$, for $i = 1 : n$.

Note that $(D_{\sigma_X})^{-1} = \text{diag}\left(\frac{1}{\sigma_i}\right)_{i=1:n}$. Let $v \in \mathbb{R}^n$, and let

$$w = (D_{\sigma_X})^{-1} v.$$

Then,

$$w^t = v^t ((D_{\sigma_X})^{-1})^t = v^t (D_{\sigma_X})^{-1}, \quad (3.34)$$

since $(D_{\sigma_X})^{-1}$ is a diagonal matrix and therefore symmetric, i.e., $((D_{\sigma_X})^{-1})^t = (D_{\sigma_X})^{-1}$.

From (3.33), (3.34), and (3.30), we find that

$$\begin{aligned} w^t \Sigma_X w &= w^t (D_{\sigma_X} \Omega_X D_{\sigma_X}) w \\ &= v^t (D_{\sigma_X})^{-1} (D_{\sigma_X} \Omega_X D_{\sigma_X}) (D_{\sigma_X})^{-1} v \\ &= v^t \Omega_X v \\ &\geq 0. \end{aligned}$$

Thus, $v^t \Omega_X v \geq 0$ for all $v \in \mathbb{R}^n$, and we conclude that Ω_X is a symmetric positive semidefinite matrix. $\square$

Question 2. Find the correlation matrix of three random variables with covariance matrix

$$\Sigma_X = \begin{pmatrix} 1 & 0.36 & -1.44 \\ 0.36 & 4 & 0.80 \\ -1.44 & 0.80 & 9 \end{pmatrix}. \quad (3.35)$$

Answer: If Ω_X is the correlation matrix of the three random variables, then

$$\Sigma_X = \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix} \Omega_X \begin{pmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{pmatrix},$$

where σ_i , $i = 1 : 3$, are the standard deviations of the three random variables, and therefore

$$\Omega_X = \begin{pmatrix} \frac{1}{\sigma_1} & 0 & 0 \\ 0 & \frac{1}{\sigma_2} & 0 \\ 0 & 0 & \frac{1}{\sigma_3} \end{pmatrix} \Sigma_X \begin{pmatrix} \frac{1}{\sigma_1} & 0 & 0 \\ 0 & \frac{1}{\sigma_2} & 0 \\ 0 & 0 & \frac{1}{\sigma_3} \end{pmatrix}.$$

Since the standard deviations of the random variables with covariance matrix Σ_X given by (3.35) are

$$\sigma_1 = \sqrt{\Sigma_X(1,1)} = 1;$$

$$\sigma_2 = \sqrt{\Sigma_X(2,2)} = 2;$$

$$\sigma_3 = \sqrt{\Sigma_X(3,3)} = 3,$$

we obtain that

$$\begin{aligned} \Omega_X &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} \Sigma_X \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0.18 & -0.48 \\ 0.18 & 1 & 0.1333 \\ -0.48 & 0.1333 & 1 \end{pmatrix}. \quad \square \end{aligned}$$

Question 3. Assume that all the entries of an $n \times n$ correlation matrix which are not on the main diagonal are equal to ρ . Find upper and lower bounds on the possible values of ρ .

Answer: Recall that a symmetric matrix with diagonal entries equal to 1 is a correlation matrix if and only if the matrix is symmetric positive semidefinite, i.e., if and only if all the eigenvalues of the matrix are nonnegative.

Let

$$\Omega = \begin{pmatrix} 1 & \rho & \dots & \rho \\ \rho & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho & \dots & \rho & 1 \end{pmatrix}.$$

We include two ways to compute the eigenvalues of Ω , which are then used to find the necessary and sufficient conditions for the matrix Ω to be a correlation matrix.

Solution 1: Note that

$$\begin{aligned}\Omega &= (1 - \rho)I + \begin{pmatrix} \rho & \rho & \cdots & \rho \\ \rho & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho & \cdots & \rho & \rho \end{pmatrix} \\ &= (1 - \rho)I + \rho M,\end{aligned}$$

where M is the $n \times n$ matrix with all entries equal to 1, and I is the $n \times n$ identity matrix.

Let λ and $v = (v_i)_{i=1:n}$ be an eigenvalue and a corresponding eigenvector of M , i.e., $Mv = \lambda v$, with $v \neq 0$.

Then, $Mv = \lambda v$ can be written as

$$\begin{cases} v_1 + v_2 + \cdots + v_n &= \lambda v_1; \\ v_1 + v_2 + \cdots + v_n &= \lambda v_2; \\ \vdots &\vdots \\ v_1 + v_2 + \cdots + v_n &= \lambda v_n, \end{cases}$$

and therefore

$$\lambda v_1 = \lambda v_2 = \cdots = \lambda v_n$$

Thus, either $\lambda = 0$, or $v_1 = v_2 = \cdots = v_n$, in which case $nv_1 = \lambda v_1$, and therefore $\lambda = n$, since $v = (v_i)_{i=1:n} \neq 0$.

In other words, the eigenvalues of M are $\lambda = 0$ and $\lambda = n$.

Note that, if $Mv = \lambda v$, then,

$$\begin{aligned}\Omega v &= (1 - \rho)v + \rho Mv = (1 - \rho)v + \rho \lambda v \\ &= (1 - \rho + \rho \lambda)v.\end{aligned}$$

Thus, $\mu = 1 - \rho + \rho \lambda$ and v are an eigenvalue and corresponding eigenvector of Ω .

Since the eigenvalues of M are $\lambda = 0$ and $\lambda = n$, it follows that the eigenvalues¹ of Ω are $\mu = 1 - \rho$, corresponding to $\lambda = 0$, and $\mu = (1 - \rho) + n\rho = 1 + (n - 1)\rho$, corresponding to $\lambda = n$.

Since Ω is a correlation matrix if and only if all its eigenvalues are nonnegative, we conclude that the matrix Ω is a correlation matrix if and only if

$$0 \leq 1 + (n - 1)\rho \quad \text{and} \quad 0 \leq 1 - \rho,$$

which is equivalent to

$$-\frac{1}{n-1} \leq \rho \leq 1. \quad (3.36)$$

Solution 2: Note that

$$\begin{aligned} \Omega &= (1 - \rho)I + \rho \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & 1 \end{pmatrix} \\ &= (1 - \rho)I + \rho \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} (1 \ \cdots \ 1) \\ &= (1 - \rho)I + \rho ww^t \\ &= (1 - \rho)I + \rho A, \end{aligned}$$

where I is the $n \times n$ identity matrix and A is the $n \times n$ matrix given by $A = ww^t$, where w is the $n \times 1$ column vector of size n with all entries equal to 1.

Recall that an $n \times n$ matrix of the form uu^t , where $u = (u_i)_{i=1:n}$ is an $n \times 1$ column vector, has an eigenvalue equal to $\sum_{i=1}^n u_i^2$ with multiplicity 1 and another eigenvalue equal to 0 with multiplicity $n - 1$.

¹The eigenvalue $1 - \rho$ has multiplicity $n - 1$, and the eigenvalue $1 + (n - 1)\rho$ has multiplicity 1; see Solution 2 of this question.

Then, the eigenvalues of the matrix A are:

$$\lambda = \sum_{i=1}^n w_i^2 = \sum_{i=1}^n 1 = n \text{ with multiplicity } 1;$$

$$\lambda = 0 \text{ with multiplicity } n - 1.$$

Note that, if λ and v are an eigenvalue and a corresponding eigenvector of A , then $Av = \lambda v$, and therefore

$$\begin{aligned}\Omega v &= (1 - \rho)v + \rho Av = (1 - \rho)v + \rho\lambda v \\ &= (1 - \rho + \rho\lambda)v.\end{aligned}$$

Thus, $1 - \rho + \rho\lambda$ and v are an eigenvalue and corresponding eigenvector of Ω . and we obtain that the matrix Ω has the following eigenvalues:

- $(1 - \rho) + n\rho = 1 + (n - 1)\rho$ with multiplicity 1;
- $1 - \rho$ with multiplicity $n - 1$.

As before, since Ω is a correlation matrix if and only if all its eigenvalues are nonnegative, we conclude that the matrix Ω is a correlation matrix if and only if

$$0 \leq 1 + (n - 1)\rho \quad \text{and} \quad 0 \leq 1 - \rho,$$

which is equivalent to

$$-\frac{1}{n-1} \leq \rho \leq 1,$$

which is the same as (3.36). $\square$

Question 4. How many eigenvalues does an $n \times n$ matrix with real entries have? How many eigenvectors?

Answer: Any $n \times n$ matrix with real entries has n eigenvalues, counted with their multiplicities; some of the eigenvalues may be complex numbers. Any $n \times n$ matrix has at most n eigenvectors.

Let A be an $n \times n$ matrix. Let λ be an eigenvalue of A with corresponding eigenvector $v \neq 0$, and let $P_A(x) = \det(xI_n - A)$ be the characteristic polynomial of A , where

I_n is the $n \times n$ identity matrix. Note that

$$\begin{aligned} Av = \lambda v, v \neq 0 &\iff (\lambda I_n - A)v = 0, v \neq 0 \\ &\iff \lambda I_n - A \text{ singular matrix} \\ &\iff \det(\lambda I_n - A) = 0 \\ &\iff P_A(\lambda) = 0. \end{aligned}$$

In other words, λ is an eigenvalue of A if and only if λ is a root of the corresponding characteristic polynomial $P_A(x)$. Since $P_A(x)$ is a polynomial of degree n , it follows from the Fundamental Theorem of Algebra that $P_A(x)$ has exactly n (complex) roots when counted with their multiplicities. We conclude that any $n \times n$ matrix has n eigenvalues, counted with their multiplicities.

An eigenvalue of multiplicity m has at least one eigenvector and at most m linearly independent corresponding eigenvectors, but it may have less than m linearly independent eigenvectors.² Thus, an $n \times n$ matrix has at most n eigenvectors, and at least as many eigenvalues as the number of distinct eigenvalues of the matrix. $\square$

Question 5. Let

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 5 \end{pmatrix}.$$

- (i) Find a 2×2 matrix M such that $M^2 = A$;
- (ii) Find a 2×2 matrix M such that $A = MM^t$.

Answer: (i) Recall that any symmetric matrix has the diagonal form

$$A = Q\Lambda Q^t, \tag{3.37}$$

²For example, the matrix $\begin{pmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{pmatrix}$ has eigenvalue 2 with multiplicity 3 and only one eigenvector, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

where Λ is the diagonal matrix whose entries on the main diagonal are the eigenvalues of A and Q is the orthogonal matrix whose columns are the corresponding eigenvectors of A of norm 1, i.e.,

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}; \quad Q = (v_1 \ v_2), \quad (3.38)$$

where $Av_1 = \lambda_1 v_1$ and $Av_2 = \lambda_2 v_2$, with $\|v_1\| = \|v_2\| = 1$.

If the matrix A has nonnegative eigenvalues, i.e., if $\lambda_1 \geq 0$ and $\lambda_2 \geq 0$, then the matrix

$$M = Q\Lambda^{1/2}Q^t \quad (3.39)$$

with

$$\Lambda^{1/2} = \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{pmatrix} \quad (3.40)$$

has the property that $M^2 = A$:

$$\begin{aligned} M^2 &= (Q\Lambda^{1/2}Q^t)(Q\Lambda^{1/2}Q^t) \\ &= Q\Lambda^{1/2}(Q^tQ)\Lambda^{1/2}Q^t \\ &= Q\Lambda^{1/2}\Lambda^{1/2}Q^t \\ &= Q\Lambda Q^t \\ &= A, \end{aligned}$$

since Q is an orthogonal matrix and therefore $Q^tQ = I$, and since, from (3.40), it follows that $\Lambda^{1/2}\Lambda^{1/2} = \Lambda$.

We now proceed to compute the eigenvalues and the eigenvectors of the matrix A . The eigenvalues of the matrix A are the roots of the characteristic polynomial $P_A(x)$

of the matrix A given by³

$$\begin{aligned} P_A(x) &= \det(xI - A) = \det \begin{pmatrix} x-2 & 2 \\ 2 & x-5 \end{pmatrix} \\ &= (x-2)(x-5) - 4 = x^2 - 7x + 6 \\ &= (x-1)(x-6). \end{aligned}$$

The roots of $P_A(x)$ are 1 and 6, and therefore the eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = 6$. The corresponding eigenvectors of norm 1 are

$$v_1 = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix} \quad \text{and} \quad v_2 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{pmatrix}.$$

For example, if $\lambda_2 = 6$, any corresponding eigenvector $v_2 = \begin{pmatrix} a \\ b \end{pmatrix} \neq 0$ is a solution to $Av = 6v$, which can be written as

$$\begin{cases} 2a - 2b = 6a \\ -2a + 5b = 6b \end{cases} \iff \begin{cases} b = -2a \\ b = -2a \end{cases}$$

Thus, any eigenvector corresponding to the eigenvalue $\lambda_2 = 6$ is of the form

$$v_2 = \begin{pmatrix} a \\ -2a \end{pmatrix} = a \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

By choosing $a = \frac{1}{\sqrt{5}}$, we obtain that an eigenvector of norm 1 corresponding to the eigenvalue $\lambda_2 = 6$ is

$$v_2 = \begin{pmatrix} \frac{1}{\sqrt{5}} \\ -\frac{2}{\sqrt{5}} \end{pmatrix}.$$

³The characteristic polynomial of the matrix A can also be obtained as follows:

$$P_A(x) = x^2 - \operatorname{tr}(A)x + \det(A) = x^2 - 7x + 6,$$

where $\operatorname{tr}(A) = 2 + 5 = 7$ and $\det(A) = 2 \cdot 5 - (-2) \cdot (-2) = 6$.

Then, it follows from (3.39) and (3.40) that the matrix M given by

$$\begin{aligned} M &= \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{pmatrix} \begin{pmatrix} \sqrt{1} & 0 \\ 0 & \sqrt{6} \end{pmatrix} \begin{pmatrix} \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{6} \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & -2 \end{pmatrix} \\ &= \frac{1}{5} \begin{pmatrix} 4 + \sqrt{6} & 2 - 2\sqrt{6} \\ 2 - 2\sqrt{6} & 1 + 4\sqrt{6} \end{pmatrix} \end{aligned}$$

has the property that $M^2 = A$.

(ii) We found that the eigenvalues of A are 1 and 6, i.e., positive. Then, A is a symmetric positive definite matrix, and therefore has a Cholesky decomposition. Recall that, if U is the Cholesky factor of the matrix A , then, by definition, $A = U^t U$. Thus, in order to find a matrix M such that $A = M M^t$, it is enough to compute the Cholesky factor

$$U = \begin{pmatrix} U(1,1) & U(1,2) \\ 0 & U(2,2) \end{pmatrix}$$

of the matrix A and let $M = U^t$.

Note that $A = U^t U$ can be written as

$$\begin{aligned} &\begin{pmatrix} A(1,1) & A(1,2) \\ A(2,1) & A(2,2) \end{pmatrix} \\ &= \begin{pmatrix} U(1,1) & 0 \\ U(1,2) & U(2,2) \end{pmatrix} \begin{pmatrix} U(1,1) & U(1,2) \\ 0 & U(2,2) \end{pmatrix}. \end{aligned} \tag{3.41}$$

In the first step of the Cholesky decomposition, the first row of U is computed as follows:

$$\begin{aligned} U(1,1) &= \sqrt{A(1,1)} = \sqrt{2}; \\ U(1,2) &= \frac{A(1,2)}{U(1,1)} = \frac{-2}{\sqrt{2}} = -\sqrt{2}; \end{aligned}$$

see also (3.41).

In the next step of the Cholesky decomposition, the entry $A(2, 2)$ of A is updated to $A(2, 2) - U(1, 2)^2 = 3$, and therefore

$$\begin{aligned} U(2, 2) &= \sqrt{A(2, 2) - (U(1, 2))^2} \\ &= \sqrt{5 - (-\sqrt{2})^2} \\ &= \sqrt{3} \end{aligned}$$

Thus, the Cholesky factor of the matrix A is

$$U = \begin{pmatrix} \sqrt{2} & -\sqrt{2} \\ 0 & \sqrt{3} \end{pmatrix},$$

and therefore the matrix $M = U^t$ given by

$$M = \begin{pmatrix} \sqrt{2} & 0 \\ -\sqrt{2} & \sqrt{3} \end{pmatrix},$$

has the property that $A = MM^t$. $\square$

Question 6. The 2×2 matrix A has eigenvalues 2 and -3 with corresponding eigenvectors $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$.

If $v = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$, find Av .

Answer: Let $\lambda_1 = 2$, $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, and $\lambda_2 = -3$, $v_2 = \begin{pmatrix} -1 \\ 3 \end{pmatrix}$. We first find constants $c_1, c_2 \in \mathbb{R}$ such that $v = c_1 v_1 + c_2 v_2$, i.e., such that

$$\begin{cases} 3 &= c_1 - c_2 \\ 1 &= 2c_1 + 3c_2 \end{cases}$$

The solution of this linear system is $c_1 = 2$ and $c_2 = -1$. Thus,

$$v = 2v_1 - v_2. \quad (3.42)$$

Since $Av_1 = \lambda_1 v_1 = 2v_1$ and $Av_2 = \lambda_2 v_2 = -3v_2$, we find from (3.42) that

$$\begin{aligned} Av &= 2Av_1 - Av_2 = 2(2v_1) - (-3v_2) \\ &= 4v_1 + 3v_2 \\ &= \begin{pmatrix} 1 \\ 17 \end{pmatrix}. \quad \square \end{aligned}$$

Question 7. Let A and B be square matrices of the same size. Show that the traces of the matrices AB and BA are equal.

Answer: Recall that, for any two square matrices A and B of the same size, the matrices AB and BA have the same characteristic polynomial, i.e.,

$$\begin{aligned} P_{AB}(x) &= \det(xI - AB) = \det(xI - BA) \\ &= P_{BA}(x), \quad \forall x \in \mathbb{R}, \end{aligned} \quad (3.43)$$

where I is the identity matrix of the same size as the matrices A and B .

Also, recall the following connection between the characteristic polynomial $P_M(x)$ of an $n \times n$ matrix M and the trace $\text{tr}(M)$ of the matrix:

$$\begin{aligned} P_M(x) &= \det(xI - M) \\ &= x^n - \text{tr}(M) + \cdots + (-1)^n \det(M). \end{aligned} \quad (3.44)$$

Since $P_{AB}(x) = P_{BA}(x)$, see (3.43), we conclude from (3.44) that

$$\text{tr}(AB) = \text{tr}(BA). \quad (3.45)$$

For completeness, we include a proof of (3.43). If the matrix B is nonsingular, then

$$xI - AB = B^{-1}(xI - BA)B,$$

and therefore

$$\begin{aligned}\det(xI - AB) &= \det(B^{-1})\det(xI - BA)\det(B) \\ &= \det(xI - BA),\end{aligned}\quad (3.46)$$

since

$$\det(B^{-1})\det(B) = \det(B^{-1}B) = \det(I) = 1.$$

If the matrix B is singular, let ϵ be a real number, and note that the matrix $B - \epsilon I$ is singular if and only if ϵ is equal to an eigenvalue of B . Since the $n \times n$ matrix B has at most n eigenvalues, it follows that, except for a finite number of values of ϵ , the matrix $B - \epsilon I$ is nonsingular, in which case we obtain from (3.46) that

$$\det(xI - A(B - \epsilon I)) = \det(xI - (B - \epsilon I)A). \quad (3.47)$$

Since both sides of (3.47) are polynomials of degree n in ϵ , and therefore continuous functions of ϵ , we can let $\epsilon \rightarrow 0$ in (3.47) and obtain that

$$\begin{aligned}\lim_{\epsilon \rightarrow 0} (\det(xI - A(B - \epsilon I))) \\ &= \lim_{\epsilon \rightarrow 0} (\det(xI - (B - \epsilon I)A)) \\ \iff \det(xI - AB) &= \det(xI - BA).\end{aligned}\quad (3.48)$$

From (3.46) and (3.48), we conclude that $\det(xI - AB) = \det(xI - BA)$ regardless of whether the matrix B is nonsingular or singular, which concludes the proof of (3.43). $\square$

Question 8. Can you find $n \times n$ matrices A and B such that

$$AB - BA = I_n,$$

where I_n is the identity matrix of size n ?

Answer: We give a proof by contradiction. If it were possible to find $n \times n$ matrices A and B such that $AB - BA = I$, then

$$\operatorname{tr}(AB - BA) = \operatorname{tr}(I_n) = n. \quad (3.49)$$

However,

$$\operatorname{tr}(AB - BA) = \operatorname{tr}(AB) - \operatorname{tr}(BA) = 0, \quad (3.50)$$

since, if A and B are square matrices, then $\operatorname{tr}(AB) = \operatorname{tr}(BA)$; cf. (3.45).

Since (3.49) and (3.50) contradict each other, we conclude that there are no matrices A and B such that $AB - BA = I$. $\square$

Question 9. A probability matrix is a matrix with non-negative entries such that the sum of the entries in each row of the matrix is equal to 1. Show that the product of two probability matrices is a probability matrix.

Answer: We first establish the following equivalent definition for a probability matrix:

The $n \times n$ matrix M is a probability matrix if and only if all the entries of M are nonnegative and

$$M\mathbf{1} = \mathbf{1}, \quad (3.51)$$

where $\mathbf{1}$ is the $n \times 1$ column vector with all entries equal to 1.

To see this, let $M = \operatorname{row}(r_j)_{j=1:n}$ be the row form of the matrix M , where r_j is an $1 \times n$ row vector, for $j = 1 : n$. The sum of all the entries in the j -th row r_j of M can be written as follows:⁴

$$\sum_{k=1}^n r_j(k) = r_j \mathbf{1}. \quad (3.52)$$

⁴Note that r_j is an $1 \times n$ vector and $\mathbf{1}$ is an $n \times 1$ vector, and therefore the expression $r_j \mathbf{1}$ from (3.52) is consistent.

Thus, the definition of a probability matrix as a matrix with the sum of the entries in each row equal to 1 can be written as

$$\begin{aligned}\sum_{k=1}^n r_j(k) &= 1, \quad \forall j = 1:n \\ \iff r_j \mathbf{1} &= 1, \quad \forall j = 1:n \\ \iff (r_j \mathbf{1})_{j=1:n} &= \mathbf{1} \\ \iff M\mathbf{1} &= \mathbf{1},\end{aligned}$$

since $M\mathbf{1} = (r_j \mathbf{1})_{j=1:n}$ if $M = \text{row}(r_j)_{j=1:n}$ is the row form of M .

In other words, we established that (3.51) is an equivalent condition for M to be a probability matrix.

Let A and B be probability matrices. Then all the entries of A and B are nonnegative, and therefore all the entries of AB are also nonnegative. From (3.51), it follows that

$$A\mathbf{1} = \mathbf{1} \quad \text{and} \quad B\mathbf{1} = \mathbf{1},$$

and therefore

$$(AB)\mathbf{1} = A(B\mathbf{1}) = A\mathbf{1} = \mathbf{1}.$$

Then, from (3.51), we conclude that AB is a probability matrix. $\square$

Question 10. Find all the values of ρ such that

$$\begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix}$$

is a correlation matrix.

Answer: Recall that a solution to this question based on was Sylvester's criterion was included in Chapter 1. We give two more solutions to this question here, one using

the Cholesky decomposition, and another one based on the definition of symmetric positive semidefinite matrices.

A symmetric matrix with diagonal entries equal to 1 is a correlation matrix if and only if the matrix is symmetric positive semidefinite. Thus, we need to find all the values of ρ such that the matrix

$$\Omega = \begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix} \quad (3.53)$$

is symmetric positive semidefinite.

Solution 1: To identify the values of ρ such that the matrix Ω is symmetric positive semidefinite, we apply the first step of the Cholesky algorithm to Ω , and obtain the following 2×2 matrix:

$$\begin{aligned} & \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} - \begin{pmatrix} 0.6 \\ -0.3 \end{pmatrix} (0.6 \ -0.3) \\ &= \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix} - \begin{pmatrix} 0.36 & -0.18 \\ -0.18 & 0.09 \end{pmatrix} \\ &= \begin{pmatrix} 0.64 & \rho + 0.18 \\ \rho + 0.18 & 0.91 \end{pmatrix}. \end{aligned}$$

Thus, the matrix Ω is symmetric positive semidefinite if and only if the matrix

$$M = \begin{pmatrix} 0.64 & \rho + 0.18 \\ \rho + 0.18 & 0.91 \end{pmatrix}$$

is symmetric positive semidefinite. Since $M(1, 1) = 0.64 > 0$, it follows that M is symmetric positive semidefinite if and only if $\det(M) \geq 0$, i.e., if and only if

$$\det(M) = 0.5824 - (\rho + 0.18)^2 \geq 0, \quad (3.54)$$

which is equivalent to

$$|\rho + 0.18| \leq \sqrt{0.5824} = 0.7632.$$

We conclude that Ω is a symmetric positive semidefinite matrix, and therefore a correlation matrix, if and only if

$$-0.7632 \leq \rho + 0.18 \leq 0.7632,$$

which can be written as

$$-0.9432 \leq \rho \leq 0.5832. \quad (3.55)$$

Note that condition (3.55) is the same as condition (1.9) obtained when solving the same question using Sylvester's Criterion; see Chapter 1.

Solution 2: By definition, the matrix Ω is symmetric positive semidefinite if and only if $x^t \Omega x \geq 0$ for all $x = (x_i)_{i=1:3} \in \mathbb{R}^3$. Note that

$$\begin{aligned} & x^t \Omega x \\ &= (x_1 \ x_2 \ x_3) \begin{pmatrix} 1 & 0.6 & -0.3 \\ 0.6 & 1 & \rho \\ -0.3 & \rho & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \\ &= x_1^2 + x_2^2 + x_3^2 + 1.2x_1x_2 - 0.6x_1x_3 + 2\rho x_2x_3. \end{aligned}$$

By completing the square, we obtain that

$$\begin{aligned} & x^t \Omega x \\ &= x_1^2 + 2x_1(0.6x_2 - 0.3x_3) + x_2^2 + x_3^2 + 2\rho x_2x_3 \\ &= (x_1 + 0.6x_2 - 0.3x_3)^2 \\ &\quad - (0.6x_2 - 0.3x_3)^2 + x_2^2 + x_3^2 + 2\rho x_2x_3 \\ &= (x_1 + 0.6x_2 - 0.3x_3)^2 \\ &\quad + 0.64x_2^2 + 2x_2x_3(\rho + 0.18) + 0.91x_3^2. \end{aligned}$$

By completing the square once again, we find that

$$\begin{aligned}
 & 0.64x_2^2 + 2x_2x_3(\rho + 0.18) + 0.91x_3^2 \\
 = & \left(0.8x_2 + x_3 \frac{\rho + 0.18}{0.8}\right)^2 \\
 & - x_3^2 \frac{(\rho + 0.18)^2}{0.64} + 0.91x_3^2 \\
 = & \left(0.8x_2 + x_3 \frac{\rho + 0.18}{0.8}\right)^2 \\
 & + \frac{x_3^2}{0.64} (0.5824 - (\rho + 0.18)^2),
 \end{aligned}$$

and therefore

$$\begin{aligned}
 & x^t \Omega x \\
 = & (x_1 + 0.6x_2 - 0.3x_3)^2 \\
 & + \left(0.8x_2 + x_3 \frac{\rho + 0.18}{0.8}\right)^2 \\
 & + \frac{x_3^2}{0.64} (0.5824 - (\rho + 0.18)^2).
 \end{aligned}$$

Thus, $x^t \Omega x \geq 0$ for all $x = (x_i)_{i=1:3} \in \mathbb{R}^3$ if and only if

$$\begin{aligned}
 & (x_1 + 0.6x_2 - 0.3x_3)^2 \\
 & + \left(0.8x_2 + x_3 \frac{\rho + 0.18}{0.8}\right)^2 \\
 & + \frac{x_3^2}{0.64} (0.5824 - (\rho + 0.18)^2) \\
 \geq & 0, \quad \forall x_1, x_2, x_3 \in \mathbb{R}.
 \end{aligned}$$

The last inequality holds if and only if

$$0.5824 - (\rho + 0.18)^2 \geq 0, \quad (3.56)$$

which is the same as (3.54).

We conclude that Ω is a correlation matrix if and only if

$$-0.9432 \leq \rho \leq 0.5832. \quad \square$$

3.3 Financial instruments: options, bonds, swaps, forwards, futures.

Question 1. The prices of three put options with strikes 40, 50, and 70, but otherwise identical, are \$10, \$20, and \$30, respectively. Is there an arbitrage opportunity present? If yes, how can you make a riskless profit?

Answer: If an arbitrage exists, it will be due to the fact that the convexity of put option values with respect to the strike price is violated.

In the plane (K, y) , the line passing through the points $(K = 40, P(40) = 10)$ and $(K = 70, P(70) = 30)$ is given by

$$y = \frac{70 - K}{30} \cdot 10 + \frac{K - 40}{30} \cdot 30. \quad (3.57)$$

The point on this line corresponding to strike 50 is obtained by substituting $K = 50$ in (3.57), and has y -coordinate equal to

$$\frac{2}{3} \cdot 10 + \frac{1}{3} \cdot 30 = \frac{50}{3}.$$

Since $P(K)$ is a strictly convex function of K , a no-arbitrage value of the put option with strike 50 should be below the line passing through the price points of the options with strikes 40 and 70. However, $P(50) = 20 > \frac{50}{3}$. Thus, the put option with strike 50 is overpriced, and an arbitrage exists.

Using a “buy low, sell high” strategy, we can take advantage of this arbitrage opportunity as follows: buy 2 put options with strike 40, buy 1 put option with strike 70, and sell 3 put options with strike 50. There is a \$10 positive cash flow when setting up this portfolio, since

$$3 \cdot \$20 - 2 \cdot \$10 - \$30 = \$10.$$

The value $V(T)$ of the portfolio at the maturity T of the options is

$$\begin{aligned} V(T) &= 2 \max(40 - S(T), 0) \\ &\quad + \max(70 - S(T), 0) \\ &\quad - 3 \max(50 - S(T), 0). \end{aligned}$$

Note that $V(T)$ is nonnegative for any value $S(T)$ of the underlying asset at T :

If $70 \leq S(T)$, then all options expire out of the money and

$$V(T) = 0.$$

If $50 \leq S(T) < 70$, then

$$V(T) = 70 - S(T) \geq 0.$$

If $40 \leq S(T) < 50$, then

$$\begin{aligned} V(T) &= (70 - S(T)) - 3(50 - S(T)) \\ &= 2S(T) - 80 \\ &\geq 0. \end{aligned}$$

If $S(T) < 40$, then

$$V(T) = 2S(T) - 80 + 2(40 - S(T)) = 0.$$

In other words, we set up a portfolio with positive cash flow at inception which does not lose money regardless of the value of the underlying asset at time T . The risk-free profit is equal to the future value at time T of the \$10 cash flow from setting up the portfolio. $\square$

Question 2. The price of a stock is \$50. In three months, it will either be \$47 or \$52, with 50% probability. How much would you pay for an at the money put? Assume for simplicity that the stock pays no dividends and that interest rates are zero.

Answer: Recall first that real world probabilities do not play any role in valuing an option in a (one period) binomial tree model. Thus, the 50% probability stated in the question is only meant to throw you off-course.

Solution 1: The value of the option is the discounted expected value of the payoff of the option in the risk-neutral probability measure. Since interest rates are zero, this can be written as

$$P(0) = p_{RN,up}P_{up} + p_{RN,down}P_{down}. \quad (3.58)$$

The up and down factors are $u = \frac{52}{50} = 1.04$ and $d = \frac{47}{50} = 0.94$, respectively. The risk-neutral probabilities of going up and down are

$$p_{RN,up} = \frac{1-d}{u-d} = 0.6; \quad p_{RN,down} = \frac{u-1}{u-d} = 0.4.$$

The ATM put pays \$3 if the stock price goes down to \$47, i.e., $P_{down} = 3$, and expires worthless if the stock price goes up to \$52, i.e., $P_{up} = 0$. From (3.58), we find that the value of the ATM put is

$$P(0) = 0.6 \cdot 0 + 0.4 \cdot 3 = 1.2. \quad (3.59)$$

In other words, you should pay at most \$1.20 for an at the money put.

Solution 2: An insightful solution can be given by setting up a hedged portfolio. The Delta of the put option is

$$\Delta_P = \frac{P_{up} - P_{down}}{S_{up} - S_{down}} = \frac{0 - 3}{52 - 47} = -0.6.$$

A portfolio which is long one ATM put and short Δ_P shares will be long the put and long 0.6 shares, and will have the same value at maturity regardless of whether the stock price goes down to \$47 or up to \$52:

$$\begin{aligned} \Pi(T) &= P(T) + 0.6S(T) \\ &= \begin{cases} 0 + 0.6 \cdot 52 = 31.20, & \text{if } S(T) = 52; \\ 3 + 0.6 \cdot 47 = 31.20, & \text{if } S(T) = 47. \end{cases} \end{aligned}$$

For no-arbitrage, the value of the portfolio at inception must be the discounted value of its payoff. Since the interest rates are zero, we obtain that $\Pi(0) = P(0) + 0.6S(0) = 31.20$, and therefore

$$P(0) = 31.20 - 0.6 \cdot 50 = 1.20,$$

which is the same value of the put, \$1.20, obtained above; see (3.59). $\square$

Question 3. A stock worth \$50 today will be worth either \$60 or \$40 in three months, with equal probability. The value of a three months at the money put on this stock is \$4. Does the value of the three months ATM put increase or decrease, and by how much, if the probability of the stock going up to \$60 were 75% and the probability of the stock going down to \$40 were 25%?

Answer: In a one period binomial tree model, the actual probabilities of the asset going up or down do not play any role in the valuation of a plain vanilla option. Thus, the value of the three months at the money put would be the same, \$4, even if the probability of the stock going up to \$60 were 75%. $\square$

Question 4. What is risk-neutral pricing?

Answer: Risk-neutral pricing, or valuation, refers to valuing derivative securities as discounted expected values of their payoffs at maturity, under the assumption that the underlying asset has lognormal distribution with a drift equal to the risk-free rate.

More precisely, if the price of the underlying asset has a lognormal distribution with volatility σ and pays dividends continuously at the rate q , then the value of a derivative security on this asset with payoff $V(T)$ at maturity T given by risk-neutral valuation is

$$V(0) = e^{-rT} E_{RN}[V(S(T))],$$

where r is the risk-free rate assumed to be constant, and the expected value is computed with respect to the log-normal random variable $S(T)$ given by

$$S(T) = S(0) e^{\left(r - q - \frac{\sigma^2}{2}\right)T + \sigma\sqrt{T}Z},$$

Risk-neutral valuation can be used for derivative securities which can be perfectly hedged dynamically using cash and the underlying asset. Plain vanilla European options, as well as European options with other payoffs at maturity (such as asset-or-nothing and cash-or-nothing options) can be priced using risk-neutrality. Risk-neutral valuation cannot be used for path dependent options such as American options, barrier options, and Asian options. $\square$

Question 5. Describe briefly how you arrive at the Black-Scholes formula.

Answer: The Black-Scholes formulas give the values of plain vanilla European put and call options on an underlying asset with lognormal distribution. Several methods for deriving the Black-Scholes formulas are:

- Risk neutral pricing: the expected value of the payoff of the option at maturity computed under the assumption that the price of the underlying asset has a lognormal distribution with drift equal to the risk free rate gives the Black-Scholes value of the option.
- Black-Scholes PDE solution: the Black-Scholes value of the option satisfies the Black-Scholes PDE with boundary conditions given by the payoff of the option at maturity. The Black-Scholes PDE is transformed into the heat PDE using a lognormal change of variables, and the closed form solution of the heat PDE is then used to derive the closed form solution of the Black-Scholes PDE, which is the Black-Scholes value of the option.

- Binomial tree model pricing: the evolution of the underlying asset is modeled using a binomial tree calibrated to converge in the limit to a lognormal distribution with drift equal to the risk free rate. For every tree, an approximate option value is obtained from the binomial tree model. The limit of these binomial tree option values as the number of time intervals in the tree goes to infinity is the Black-Scholes value of the option.

Note that twelve different ways to derive the Black-Scholes formula can be found in Wilmott [4]. $\square$

Question 6. How much should a three months at the money put on an asset with spot price \$40 and volatility 30% be worth? Assume, for simplicity, that interest rates are zero and that the asset does not pay dividends.

Answer: The following approximation for the value of an at the money put option on a non dividend paying underlying asset and assuming zero risk-free interest rates is easy to estimate and very accurate if the total variance is small (e.g., if $\sigma^2 T \leq 0.25$):

$$P_{ATM} \approx 0.4\sigma S_0 \sqrt{T}; \quad (3.60)$$

see Stefanica [3] for a derivation of formula (3.60).

For $S_0 = 40$, $\sigma = 0.3$, and $T = \frac{1}{4}$, we obtain that the value of the at the money put is approximately 2.40.

For comparison purposes, note that the value of the put option computed using the Black-Scholes formula would be 2.3914; the approximate formula (3.60) is very accurate in this case. $\square$

Question 7. If the price of a stock doubles in one day, by how much will the value of a call option on this stock change?

Answer: The value of a deep in the money call on a non dividend paying asset can be approximated, e.g., by using

the Put-Call parity, as $C \approx S - Ke^{-rT}$, where K and T are the strike and the maturity of the option, and r is the constant risk free rate. Thus, if the spot price S doubles, the call option will be even deeper in the money, and therefore its value will be approximately $2S - Ke^{-rT}$. In other words, the value of deep in the money calls roughly doubles if the spot price doubles.

If the option is around at the money, the percentage change generated by the doubling of the stock price is about one order of magnitude larger since the option will become deep in the money.

If the option is deep out of the money, then it trades for fraction of cents. The doubling of the spot price would result in changing of the price of the option by several orders of magnitude.

As a numerical example, consider a six months call option with strike 20 on a non dividend paying underlying asset with volatility 25%. Assume that the risk free rate is constant at 5%. The Black-Scholes values of the call option corresponding to several spot prices of the underlying asset can be found below:

Spot Price	Option Price
10	0.000045
20	1.65
40	20.49
80	60.49
400	380.49
800	780.49

If the call option is deep out of the money and the spot price doubles from \$10 to \$20, the value of the call increases from \$0.000045 to \$1.65, i.e., by more than four orders of magnitude.

If the call option is at the money and the spot price doubles from \$20 to \$40, the value of the call increases

from \$1.65 to \$20.49, i.e., more than tenfold.

If the call option is deep in the money, and the spot price doubles from \$40 to \$80, the value of the call increases from \$20.49 to \$60.49, i.e., by a factor of 2.95; if the call option is even deeper in the money⁵ and the spot price doubles from \$400 to \$800, the value of the call increases from \$380.49 to \$780.49, i.e., the call approximately doubles in value.

Moreover, if the call option is deep in the money, its value is very close to $S - Ke^{-rT}$, i.e., the value of the spot price of the underlying asset minus the present value of the strike. For all the spot prices greater than \$40, the estimate $C \approx S - Ke^{-rT}$ is very accurate. Thus, if the call option is at the money and the spot price doubles, the value of the call option increases by the same amount as the increase in the spot price. For example, if the spot price doubles from \$40 to \$80, the value of the call increases by \$40, from \$20.49 to \$60.49, which is exactly the increase in the spot price. $\square$

Question 8. What are the smallest and largest values that Delta can take?

Answer: Assume, for simplicity, that the underlying asset does not pay dividends.

The Delta (Δ) of a long position in a plain vanilla call option is between 0 and 1 (and therefore the Delta of a short plain vanilla call position is between -1 and 0). The Delta of a long call position increases with the spot price of the underlying asset, and goes from 0, when the asset is worthless (i.e., when the call option is deep out of the money), to 1, when the spot price of the asset is very large (i.e., when the call option is deep in the money).

⁵Of course, call options with strike ten times smaller than the spot price of the underlying asset never occur in practice; this part of the example is for illustration purposes.

The Delta of a long position in a plain vanilla put option is between -1 and 0 ; the Delta of a short position in a plain vanilla put option is between 0 and 1 . The Delta of a long put position also increases with the spot price of the underlying asset, and goes from -1 , when the asset is worthless (i.e., when the put option is deep in the money), to 0 , when the spot price of the asset is very large (i.e., when the put option is deep out of the money).

Note that, from the Put-Call parity, i.e., $C - P = S - Ke^{-rT}$, it follows that $\Delta(P) = \Delta(C) - 1$, which is consistent with the bounds above. $\square$

Question 9. What is the Delta of an at-the-money call? What is the Delta of an at-the-money put?

Answer: The Delta of an at-the-money call is approximately 0.5 ; the Delta of an at-the-money put is approximately -0.5 .

Assume, for simplicity, a Black-Scholes framework with zero risk-free rates and an underlying asset paying no dividends. Then, $\Delta(C_{BS}) = N(d_1)$, where $N(x)$ denotes the cumulative distribution of the standard normal variable and

$$d_1 = \frac{\ln\left(\frac{S}{K}\right) + \left(r - q + \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}} = \frac{\sigma\sqrt{T}}{2} \quad (3.61)$$

for $K = S$ and for $r = q = 0$.

The linear Taylor approximation of $N(x)$ around 0 is

$$N(x) \approx 0.5 + \frac{x}{\sqrt{2\pi}} \approx 0.5 + 0.4x, \quad (3.62)$$

since $\frac{1}{\sqrt{2\pi}} = 0.3989 \approx 0.4$. Thus,

$$\Delta(C_{BS}) = N(d_1) = N\left(\frac{\sigma\sqrt{T}}{2}\right) \approx 0.5 + 0.2\sigma\sqrt{T}.$$

This is roughly estimated as $\Delta(C_{BS}) \approx 0.5$, since $0.2\sigma\sqrt{T}$ is small for most options (e.g., $0.2\sigma\sqrt{T} \leq 0.1$ for volatility less than 50% and for maturity less than one year).

For put options, $\Delta(P_{BS}) = -N(-d_1)$, where $d_1 = \frac{\sigma\sqrt{T}}{2}$; cf. (3.61). From (3.62), it follows that

$$\Delta(P_{BS}) = -N\left(-\frac{\sigma\sqrt{T}}{2}\right) \approx -0.5 + 0.2\sigma\sqrt{T},$$

which, using the same rationale outlined above, is often stated as $\Delta(P_{BS}) \approx -0.5$. $\square$

Question 10. What is the Put–Call parity? How do you prove it?

Answer: The Put–Call parity is a model independent no-arbitrage relationship between the prices of European call and put options with the same strike and maturity.

In a nutshell, the Put–Call parity states that being long a call option and short a put option on the same underlying asset, and with the same strike and maturity, is the same as being long a forward contract on the asset, with the same maturity as the maturity of the options, and with delivery price equal to the strike of the options. Equivalently, being long a call and short a put is the same as being long one unit of the underlying asset (for non dividend paying assets) and short the present value of the strike of the options.

More precisely, if $C(t)$ and $P(t)$ are the values at time t of a European call and put option with maturity T and strike K , on the same non dividend paying asset with spot price $S(t)$, then the Put–Call parity states that

$$C(t) - P(t) = S(t) - Ke^{-r(T-t)}, \quad (3.63)$$

where r denotes the risk-free rate, assumed to be constant.

If the underlying asset pays dividends continuously at the rate q , the Put-Call parity has the form

$$C(t) - P(t) = S(t)e^{-q(T-t)} - Ke^{-r(T-t)}. \quad (3.64)$$

For simplicity, we restrict our attention to non dividend paying underlying assets and prove formula (3.63).

Consider a portfolio made of the following assets:

- long 1 put option;
- long 1 unit of the underlying asset;
- short 1 call option.

The value of the portfolio at time t is

$$V_{portfolio}(t) = P(t) + S(t) - C(t). \quad (3.65)$$

The values of the call and put option at maturity are

$$\begin{aligned} C(T) &= \max(S(T) - K, 0); \\ P(T) &= \max(K - S(T), 0). \end{aligned}$$

Then, regardless of whether $S(T) < K$ or $S(T) \geq K$, the value of the portfolio at time T will be equal to K :

If $S(T) < K$, then $P(T) = K - S(T)$ and $C(T) = 0$, and therefore

$$\begin{aligned} V_{portfolio}(T) &= P(T) + S(T) - C(T) \\ &= (K - S(T)) + S(T) - 0 \\ &= K. \end{aligned} \quad (3.66)$$

If $S(T) \geq K$, then $P(T) = 0$ and $C(T) = S(T) - K$, and therefore

$$\begin{aligned} V_{portfolio}(T) &= P(T) + S(T) - C(T) \\ &= 0 + S(T) - (S(T) - K) \\ &= K. \end{aligned} \quad (3.67)$$

From (3.66) and (3.67), we find that

$$V_{portfolio}(T) = P(T) + S(T) - C(T) = K, \quad (3.68)$$

regardless of the value $S(T)$ of the underlying asset at the maturity of the option.

Then, for no-arbitrage, the value of the portfolio at time t must be equal to the present value of K at time t , i.e.,

$$V_{\text{portfolio}}(t) = Ke^{-r(T-t)}. \quad (3.69)$$

From (3.65) and (3.69), we obtain that

$$P(t) + S(t) - C(t) = Ke^{-r(T-t)}.$$

This can be written as

$$C(t) - P(t) = S(t) - Ke^{-r(T-t)},$$

which is the Put-Call parity formula (3.63).

The Put-Call parity formula (3.64) corresponding to an underlying asset paying dividends continuously at the rate q can be obtained similarly using a portfolio with a long put position, a short call position, and a long position in $e^{-q(T-t)}$ units of the underlying asset. All dividends paid by the long asset position between time t and time T are used to purchase additional fractions of the asset. Doing so continuously results in an asset position at time T equal to long one unit of the asset. Thus, the value of the portfolio at time T will be equal to K regardless of the value of $S(T)$, and the Put-Call parity formula (3.64) for underlying assets paying dividends continuously can be obtained as before. $\square$

Question 11. Show that the time value of a European call option is highest at the money.

Answer: The time value of a call option is the difference between the value $C(S)$ of the option and the intrinsic value $\max(S - K, 0)$ of the call option. In other words, the time value of the option is

$$C(S) - \max(S - K, 0).$$

We want to establish that the time value of the option is highest at the money, i.e., when $S = K$. To do so, we show that the function

$$f(S) = C(S) - \max(S - K, 0)$$

attains its maximum for $S = K$.

Note that⁶

$$f(S) = \begin{cases} C(S), & \text{if } S \leq K; \\ C(S) - S + K, & \text{if } S > K. \end{cases}$$

For $S \leq K$, the function $f(S)$ is the value of a call with strike K , and therefore is increasing.

For $S > K$, we find that

$$f'(S) = \Delta(C) - 1 < 0,$$

since the Delta of a call option is always less than 1,⁷ and therefore the function $f(S)$ is decreasing.

We conclude that $f(S)$ has a global maximum point at $S = K$, which is what we wanted to show.

Note that a similar reasoning shows that the time value of a put option, given by

$$P(S) - \max(K - S, 0),$$

is largest at the money, i.e., for $S = K$. If $g(S) = P(S) - \max(K - S, 0)$, then

$$g(S) = \begin{cases} P(S) - K + S, & \text{if } S \leq K; \\ P(S), & \text{if } S > K. \end{cases}$$

⁶The function $f(S)$ is a continuous function, but it is not differentiable at $S = K$.

⁷The Delta of a long call position is an increasing function going from 0 when the spot price of the underlying asset is 0 and the call option is deep out of the money, to 1 when the spot price of the underlying asset goes to ∞ and the call option is deep in the money. For example, in the Black-Scholes model,

$$\Delta(C_{BS}) = e^{-qT} N(d_1) < N(d_1) < 1.$$

For $S \leq K$,

$$g'(S) = \Delta(P) + 1 > 0,$$

and therefore the function $g(S)$ is increasing, since the Delta of a put option is between -1 (when the put is deep in the money) and 0 (when the put is deep out of the money), and therefore is always greater than -1 .

For $S > K$, the function $g(S)$ is the value of a put with strike K , and therefore is decreasing.

Thus, $g(S)$ has a global maximum point at $S = K$, and therefore the time value of the put option is largest at the money. $\square$

Question 12. What is implied volatility? What is a volatility smile? How about a volatility skew?

Answer: By definition, implied volatility is the unique value of the volatility parameter σ from the lognormal model for the evolution of the price of an underlying that makes the Black-Scholes value of an option equal to the market price of the option. Implied volatility exists and is unique⁸ for any arbitrage-free market value of the option.

On the same asset, prices of options with multiple strikes and maturities are quoted, and implied volatilities can be computed for each of these options. If the price of the asset had a lognormal distribution as assumed in the Black-Scholes model, then the resulting plots of implied volatility vs strike for the same maturity should be flat. In practice, they are not flat, and are often shaped as “smiles” or “skews”.

An implied volatility smile occurs when the implied volatilities of deep in the money options and of deep out of the money options are higher than the implied volatilities

⁸The uniqueness of the implied volatility comes from the fact that the Black-Scholes value of the option is a strictly increasing function of the volatility parameter (or, equivalently, the vega of the option is strictly positive).

of options close to at the money. Volatility smiles are typical for currency options.

An implied volatility skew occurs when the implied volatilities of options with large strikes are lower than the implied volatilities of at the money options (reverse skew), or when the implied volatilities of options with small strikes are lower than the implied volatilities of at the money options (forward skew). Reverse skews are typical for long dated equity and index options. Forward skews are typical for commodities options. $\square$

Question 13. What is the Gamma of an option? Why is it preferable to have small Gamma? Why is the Gamma of plain vanilla options positive?

Answer: The Gamma (Γ) of an option measures the sensitivity of the Delta of the option with respect to the price of the underlying asset, i.e.,

$$\Gamma(V) = \frac{\partial \Delta}{\partial S} = \frac{\partial^2 V}{\partial S^2},$$

where V denotes the value of the option.

It is often important to immunize a portfolio with respect to changes in the price of the underlying asset, i.e., to make the portfolio Delta-neutral. A portfolio with small Gamma would need to be rebalanced less often in order to be kept Delta-neutral, since the change in the Delta of the portfolio is proportional to Gamma for small changes in the value of the underlying asset. Thus, a Delta-neutral and Gamma-neutral portfolio is well hedged against small changes in the price of the underlying asset (although not against jumps in the price of the underlying asset).

The Delta of plain vanilla options (calls or puts) increases as the spot price of the underlying asset increases. Thus, Gamma is positive, since Gamma is the rate of

change of Delta. Moreover, Gamma is asymptotically going to 0 for deep out of the money options and for deep in the money options, and the highest value of Gamma corresponds to options with strike close to the spot price (at the money options). $\square$

Question 14. When are a European call and a European put worth the same? (The options are written on the same asset and have the same strike and maturity.) What is the intuition behind this result?

Answer: Recall from the Put–Call parity that

$$C - P = Se^{-qT} - Ke^{-rT}, \quad (3.70)$$

where C and P are the values of a call and put option, respectively, with strike K and maturity T on an underlying asset with spot price S and paying dividends continuously at rate q . If $C = P$, it follows from (3.70) that $K = Se^{(r-q)T}$. Since the forward price of the underlying asset is $F = Se^{(r-q)T}$, we conclude that a call and a put are worth the same if their strike is equal to the forward price of the asset; these options are called at-the-money-forward options.

Note that this result is independent of any assumption on the evolution of the price of the underlying asset, since Put–Call parity is model independent.

The fact that an at-the-money-forward call and an at-the-money-forward put are worth the same may seem counterintuitive at first glance: call options have unlimited upside since their payoff at maturity,

$$C(T) = \max(S(T) - K, 0),$$

can be infinitely large, while put options have limited upside since their payoff at maturity,

$$P(T) = \max(K - S(T), 0),$$

is bounded by the strike price, i.e., $P(T) \leq K$.

However, the value of an option is equal to the risk-neutral expected value of the option payoff at maturity. In every model for the evolution of the price of the underlying asset (including in the geometric Brownian motion model underlying the Black-Scholes framework), the probability density of the underlying asset at maturity decreases exponentially for large values of the spot price. This renders the expected value of large payoffs negligible, and makes it possible for at-the-money-forward put options to be worth the same as at-the-money-forward call options. $\square$

Question 15. What is the two year volatility of an asset with 30% six months volatility?

Answer: Asset volatility scales with the square root of time: if σ denotes the annualized volatility of an asset, the volatility $\sigma(t)$ of the asset over a time horizon t is given by $\sigma(t) = \sigma\sqrt{t}$. Then,

$$\frac{\sigma(t_2)}{\sigma(t_1)} = \sqrt{\frac{t_2}{t_1}}.$$

For $t_2 = 2$, $t_1 = 0.5$, and $\sigma(t_1) = \sigma(0.5) = 0.3$, we find that $\sigma(t_2) = \sigma(2) = 0.6$, i.e., the two year volatility of the asset is 60%. $\square$

Question 16. How do you value an interest rate swap?

Answer: For valuation purposes only, add payments at maturity equal to the notional of the swap both for the fixed leg and for the floating leg of the swap. Then, the value of the swap for the party receiving fixed payments is $V_{\text{swap}} = V_{\text{fix}} - V_{\text{float}}$, where V_{fix} is the value of a coupon bond with coupon rate equal to the fixed rate of the swap, and V_{float} is the value of an instrument making

the floating payments of the swap, plus a payment equal to the notional at maturity.

The value V_{fix} of the fixed rate coupon bond is the sum of the present values of its future payments discounted using risk-free zero rates (most frequently, LIBOR rates).

To compute V_{float} , note that, right after a payment is made, the remaining payments of the floating leg (including the notional at maturity) are equivalent to rolling over the notional at the prevailing zero rates until maturity. Thus, the value of all the floating payments on a payment date for the swap is equal to the notional. We conclude that V_{float} is the present value of the next floating payment (which was determined at the prior swap payment date) plus the notional.

To illustrate swap valuation with an example, consider a 19-month semiannual swap on a \$10 million notional with 3% fixed rate and paying semiannually compounded LIBOR. The next floating payment that will be made in one month is \$125,000 (and was determined five months ago, at the previous cash flow date).

The cash flow dates of the swap are 1 month, 7 months, 13 months, and 19 months. The value of the swap for the party receiving fixed payments is $V_{swap} = V_{fix} - V_{float}$. The value of the 3% semiannual coupon bond corresponding to the fixed leg of the swap is

$$\begin{aligned}
 V_{fix} = & 150,000 \cdot \text{disc} \left(\frac{1}{12} \right) \\
 & + 150,000 \cdot \text{disc} \left(\frac{7}{12} \right) \\
 & + 150,000 \cdot \text{disc} \left(\frac{13}{12} \right) \\
 & + 10,150,000 \cdot \text{disc} \left(\frac{19}{12} \right), \quad (3.71)
 \end{aligned}$$

where $\text{disc}(t)$ denotes the discount factor corresponding

to time t . For example, if the discount factors are given in terms of the semiannually compounded LIBOR rate $\text{LIBOR}(t)$, then

$$\text{disc}(t) = \left(1 + \frac{\text{LIBOR}(t)}{2}\right)^{-2t}.$$

On the next cash flow date, i.e., in one month, the floating leg of the swap is equal to the value of the next floating payment, \$125,000, plus the \$10 million notional, i.e., \$10,125,000. Thus, the value of the floating leg of the swap today is the present value of \$10,125,000 in one month, i.e.,

$$V_{float} = 10,125,000 \cdot \text{disc}\left(\frac{1}{12}\right). \quad (3.72)$$

The value of the swap for the party receiving fixed payments is $V_{swap} = V_{fix} - V_{float}$, where V_{fix} and V_{float} are given by (3.71) and (3.72), respectively. $\square$

Question 17. By how much will the price of a ten year zero coupon bond change if the yield increases by ten basis points?

Answer: The first order approximation of the change ΔB in the bond price in terms of the change Δy in the bond yield is

$$\Delta B \approx \frac{\partial B}{\partial y} \Delta y = -DB \Delta y, \quad (3.73)$$

where $D = -\frac{1}{B} \frac{\partial B}{\partial y}$ is the duration of the bond. Thus,

$$\frac{\Delta B}{B} \approx -D \Delta y. \quad (3.74)$$

Note that $D = 10$, since the duration of a zero coupon bond is equal to the maturity of the bond. Moreover, the change in the bond yield is $\Delta y = 0.001$, since $1\% = 100$ basis points (bps), and therefore $10 \text{ bps} = 0.1\% = 0.001$.

From (3.74), we find that the percentage change in the value of the bond can be estimated as follows:

$$\frac{\Delta B}{B} \approx -10 \times 0.001 = -0.01.$$

In other words, the price of the ten year zero-coupon bond decreases by 1% if the yield increases by ten basis points. $\square$

Question 18. A five year bond with 3.5 years duration is worth 102. What is the value of the bond if the yield decreases by fifty basis points?

Answer: The value of the bond will increase, since the yield of the bond decreases. More precisely, recall from (3.73) that

$$\Delta B \approx -DB\Delta y. \quad (3.75)$$

For $B = 102$, $D = 3.5$ and $\Delta y = -0.005$ ($1\% = 100$ bps, and therefore 50 bps $= 0.5\% = 0.005$), we find from (3.75) that $\Delta B \approx 1.785$. Thus, the new value of the bond is $B + \Delta B \approx 103.785$. $\square$

Question 19. What is a forward contract? What is the forward price?

Answer: A forward contract is an agreement between two parties in which one party (the long position) agrees to buy a specified quantity of the underlying asset from the other party (the short position) at a given time in the future and for a price, called the forward price, that is agreed upon at the inception of the forward contract. The forward price is chosen such that the forward contract has value 0 at inception.⁹

If the underlying asset has spot price S_0 and pays dividends continuously at rate q , the forward price for a for-

⁹Note that the forward price is not the price of the forward.

ward contract maturing at T is

$$F = S_0 e^{(r-q)T},$$

where r is the risk-free rate between 0 and T . $\square$

Question 20. What is the forward price for treasury futures contracts? What is the forward price for commodities futures contracts?

Answer: A short position in a forward contract (i.e., selling a forward contract) is perfectly hedged by buying one unit of the underlying asset and holding that position until the maturity of the forward contract. The forward price is the future value at the maturity of the forward contract of the cost of buying one unit of the underlying asset.

For a treasury futures contract, buying the underlying treasury bond generates a positive cash flow from receiving all the bond coupon payments until the maturity of the forward contract. If S_0 is the spot price of the underlying treasury bond and if C is the present value of all the coupon payments received during the life of the futures contract, then the forward price F of a treasury futures contract with maturity T is the future value at time T of $S_0 - C$, i.e., $F = (S_0 - C)e^{rT}$.

For a commodities futures contract, buying the underlying commodity incurs storage costs; for example, for a gold futures contract, buying the underlying gold would require storing the gold safely. If S_0 is the spot price of the underlying commodity and if C is the present value of all the storage costs, then the forward price F of a commodities futures contract with maturity T is the future value at time T of $S_0 + C$, i.e., $F = (S_0 + C)e^{rT}$, where r denotes the risk-free rate between 0 and T . $\square$

Question 21. What is a Eurodollar futures contract?

Answer: A Eurodollar is a dollar deposit in a bank outside the United States. The Eurodollar rate is the interest rate earned on Eurodollars deposited by one bank with another bank (and is very close to LIBOR for short term maturities).

A Eurodollar futures contract is a futures contract on a Eurodollar rate, and deliveries are for up to ten years in the future.

For example, a three-month Eurodollar futures contract is a futures contract on the three month (90-day) Eurodollar rate. The start date is the third Wednesday of the delivery month (March, June, September, December). □

Question 22. What are the most important differences between forward contracts and futures contracts?

Answer: The main differences between the ways forward and futures contracts are structured, settled, and traded are:

- Futures contracts trade on exchanges and have standard features, while forward contracts are over-the-counter instruments.
- Futures are marked to market and settled in a margin account on a daily basis, while forward contracts are not settled before maturity.
- Futures carry almost no credit and counterparty risk, since they are settled daily, while entering into a forward contract carries some credit risk.
- Futures have a range of delivery dates, while forward contracts have a specified delivery date.
- Futures contracts require the delivery of the underlying asset for the futures price, while forward contracts can be settled in cash at maturity, without the delivery of a physical asset. □

Question 23. What is the ten-day 99% VaR of a portfolio with a five-day 98% VaR of \$10 million?

Answer: If we assume normally distributed short term portfolio returns, the VaR (Value at Risk) of a portfolio is proportional to the square root of the time horizon. More precisely, if the N day $C\%$ VaR of the portfolio is denoted by $\text{VaR}(N, C)$, where N is the number of days in the time horizon and C is the confidence level, then

$$\text{VaR}(N, C) \approx \sigma_V z_C \sqrt{\frac{N}{252}} V(0), \quad (3.76)$$

where, σ_V is the (annualized) standard deviation of the rate of return of the portfolio, z_C is the z -score of the standard normal distribution corresponding to C , i.e., $P(Z \leq z_C) = C$, and $V(0)$ is the current value of the portfolio.

From (3.76), it follows that

$$\text{VaR}(10 \text{ days}, 99\%) \approx \frac{z_{99} \sqrt{10}}{z_{98} \sqrt{5}} \text{VaR}(5 \text{ days}, 98\%).$$

For $\text{VaR}(5 \text{ days}, 98\%) = \$10,000,000$, and since $z_{99} \approx 2.326348$ and $z_{98} \approx 2.053749$, we obtain that

$$\text{VaR}(10 \text{ days}, 99\%) \approx \$16,019,255. \quad \square$$

3.4 C++. Data structures.

Question 1. How do you declare an array?

Answer: An array can be declared either on stack, or on heap.

```
//created on stack, uninitialized  
T identifier[size];
```

```
//created on stack, initialized  
T identifier[] = initializer_list;
```

```
//created on heap, uninitialized  
T* identifier = new T[size];
```

Example:

```
int foo[3];  
int bar[] = {1,2,3};  
int* baz = new int[3];
```

Question 2. How do you get the address of a variable?

Answer: Use the ampersand before the name of the variable, e.g.

```
T var;  
T* ptr = &var
```

Example:

```
int foo = 1;  
int* foo_ptr = &foo;
```

Question 3. How do you declare an array of pointers?

Answer: The same way as declaring an array, but making the type, T, a pointer:

```
T* identifier[size];  
T* identifier[] = initializer_list;  
T** identifier = new T[size];
```

Example:

```
int a = 1; int b = 2; int c = 3;  
int* foo[3];  
int* bar[] = {&a, &b, &c};  
int** baz = new int*[3];
```

Question 4. How do you declare a const pointer, a pointer to a const and a const pointer to a const?

Answer:

```
//pointer to a read only variable  
const T* identifier;  
T const* identifier;
```

```
//read only pointer to a variable  
T *const identifier = rvalue;
```

```
//read only pointer to a read only variable  
const T *const identifier = rvalue;  
T const *const identifier = rvalue;
```

Example:

```
//read only variables  
const int a = 2; const int b = 2;  
int c = 1;
```

```
//pointer to a read only b  
int const* foo_two;  
foo_two = &a; foo_two = &b;
```

```
//pointer to read only a
```

```
const int* foo;  
foo = &a; foo = &b;  
  
//read only pointer to c  
//it needs to be initialized  
int *const bar = &c;  
  
//read only pointer to read only a  
//it needs to be initialized  
const int *const baz = &a;
```

Question 5. How do you declare a dynamic array?

Answer:

```
T* identifier = new T[size];  
T* identifier = nullptr;  
T* identifier;
```

```
delete[] identifier;
```

Example:

```
int *foo = new int[4];  
int *bar = nullptr;  
bar = new int[4];
```

Question 6. What is the general form for a function signature?

Answer:

```
return_type function_name(parameter_list);
```

Example:

```
int my_sum(int a, int b);
```